

PREM TIWARI

Software Engineer II at SmartSheet, Bengaluru, Karnataka

+91 99936 34163 | premxivfeb@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [CodeChef](#)

Professional Summary

Results-driven **SDE** with expertise in scalable **data pipelines**, **real-time processing**, and **IoT data ingestion** using protocols like **OPC-UA** and **MQTT**, alongside technologies like **Kafka**, **Redis**, **Spark**, and **AWS**. Currently at **SmartSheet**, delivering **cloud-native**, high-performance energy data solutions. Skilled in **Python**, **PySpark**, **Delta Lake**, **ETL**, **Data Warehousing**, **Data Modeling** and **Databricks**, with a focus on **optimizing large-scale workflows**. Strong team player, passionate about **collaborative problem-solving** and **building efficient, maintainable, and impactful data systems**.

Skills

Programming & Core Skills: Python, C++, Java, OOPs, DSA, Unit Testing, Mocking, Feature Engineering, Data Cleaning Algorithms, Machine Learning, Visualization, Low-Level Design (LLD), High-Level Design (HLD), Excel

Data Engineering, Cloud & Databases: AWS (EC2, S3, Lambda, SNS, SQS, Kinesis, CloudWatch, DynamoDB, Glue), Databricks, Snowflake, Airflow, Airbyte, DBT, Spark, Kafka, Redis, ETL, Data Warehousing, Data Modeling, OpenSearch, Logstash, Kinesis Autoscaling, CloudWatch Logs, PostgreSQL, MongoDB, DynamoDB, Redis

DevOps, CI/CD & Tools: Docker, Kubernetes, Git, GitPipeline/Actions, Jenkins, Terraform, Terragrunt, Ansible, MLflow, SonarQube, Postman, Grafana, Databricks Asset Bundles.

Work Experience

Software Engineer II / Smartsheet

Oct 2025 - Present

- Databricks Cost Optimization:** Migrated Databricks multi-task streaming jobs from all-purpose to job clusters, reducing compute costs by **\$200K+** annually.
- Streaming Platform (Medallion Architecture):** Built an end-to-end streaming pipeline (Airbyte → S3 → Databricks Auto Loader → Spark Structured Streaming → Gold tables) with medallion architecture for scalable, reliable data processing.
- CI/CD for Databricks Workflows:** Supported CI/CD adoption via **Databricks Asset Bundles**, enabling automated and consistent job deployments across environments.
- Real-Time Observability Pipeline:** Built and supports a high-availability log ingestion pipeline (CloudWatch → Kinesis → Logstash → OpenSearch) enabling sub-second search and **zero-data-loss** ingestion at scale.
- Kinesis Autoscaling & Logstash Reliability:** Led Kinesis stream autoscaling and Logstash pod scaling improvements to sustain stable throughput during peak traffic.
- Legacy Pipeline Reliability:** Improved legacy ingestion pipelines (Airbyte → S3 → Airflow → Snowflake) through bug fixes in ingestion logic and scheduling, reducing daily failures by **~40% in on-call**.

Data Engineer / Vestas

Feb 2024 – Sep 2025

- File Download API:** Developed an API to fetch file metadata from **DocumentDB** and retrieve files from **AWS S3** for download.
- Data Ingestion & Processing:** Developed and optimized pipelines in **Databricks**, **PySpark** to onboard and process **solar & wind farm** data, reducing **processing time by 70%**.
- Event Alarm System:** Designed real-time alarms using **Kafka**, **Redis**, **Kinesis**, improving event detection latency by 50%.
- Real-time Data Pipeline:** Built OPC UA-based ingestion pipeline to AWS Cloud, enhancing data **availability by 96%**.
- Microservices & Cost Optimization:** Containerized Databricks jobs with **Docker**, **Kubernetes**, cutting infrastructure costs by **\$4,000/month**.
- EC2 Migration & Scaling:** Optimized workloads with containerized EC2, reducing costs by **40%** and improving performance.
- POC for Migration & Cost Savings:** Led **POCs** for new pipelines, enhancing efficiency and ensuring smooth transitions.
- Job Migration & Testing:** Upgraded Databricks jobs, ensuring compatibility and **40%** performance improvement.

Assistant Data Engineer / Vestas

Feb 2023 - Jan 2024

- Event Data Consolidation:** Engineered event tracking for turbines, ensuring accurate event logging, created **LLD and HLD**.
- API Polling for Signal Data:** Automated real-time data ingestion using **API Pollers**, increasing data collection efficiency by **100%**.
- Latency Tracking:** Built a monitoring system to optimize **API response times**, reducing latency by **36%**.
- ML Model Automation:** Automated model registration in **AWS S3**, cutting manual effort by **12 hours/week**.
- MLflow Dashboard:** Developed a comparison **dashboard** for ML model accuracy.
- Databricks Compatibility Testing:** Ensured seamless **Databricks version upgrades** without pipeline failures, upgrade more than 50 jobs.
- Forecasting Model Testing:** Enhanced **model accuracy** through validation and Unit testing coverage by **95%**.
- Data Preprocessing for ML:** Optimized power curve **ML training**, improving accuracy by **7%**.

Education

Jabalpur Engineering College, Jabalpur, M.P.

Bachelor of Technology in Information Technology

July 2019 – April 2023

7.77 CGPA

Achievement

CodeChef : 5 Star, Global Rank 8, College Rank 1st (Starters 70).

LeetCode and GFG: Solved 450+ problems.

TCS_CodeVita: Round 2 Global Rank: 83rd.

GFG Job-A-Thon 6: Global Rank: 76th.

Snowflake Certifications: Collaboration, Marketplace | Data Application | Data Lake | Data Warehousing | Data Engineering

Projects

OOPs and Machine Code Projects / C++

Sept 2023

- Completed workshops on advanced inheritance, machine code, OOPs, C++, GUI and multithreading.

Live Streaming Video Chat App / Python, Socket

Feb 2023

- Built a real-time video chat application using Python, enabling seamless live video communication.